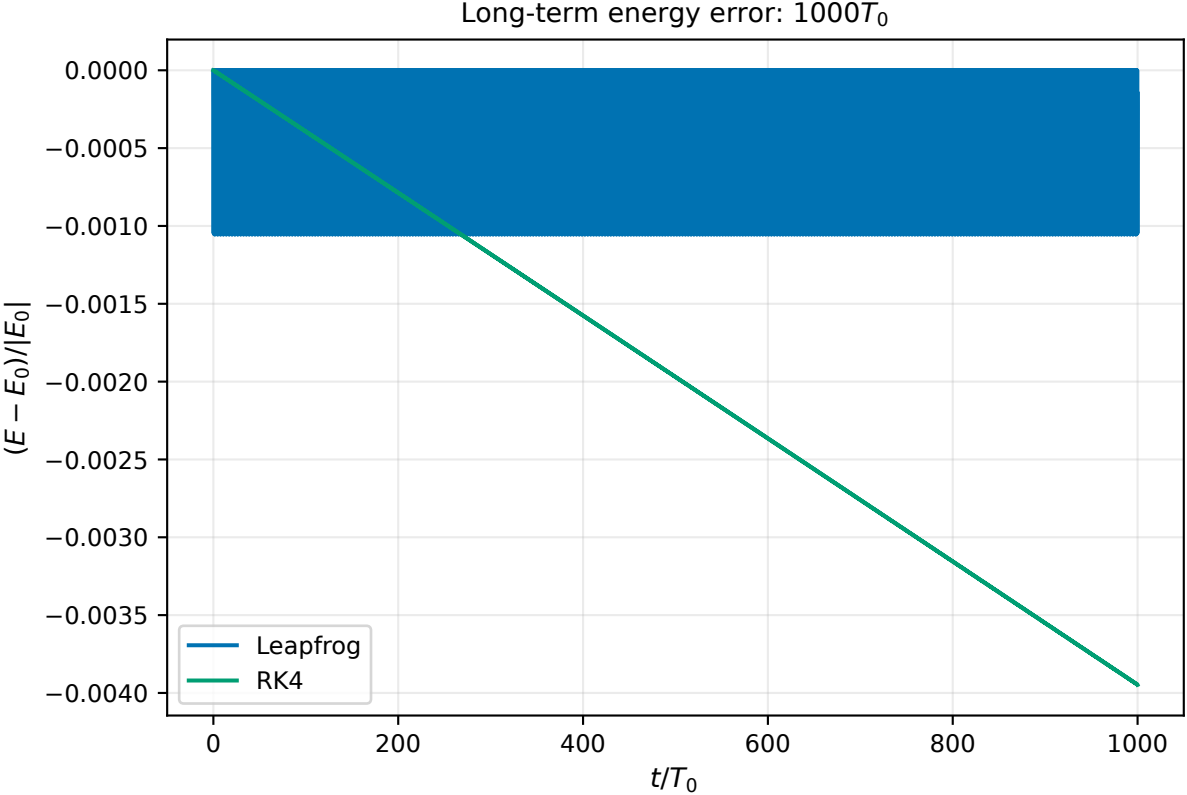

Long-term energy error: $1000T_0$
0.0000
−0.0005
−0.0010
−0.0015
−0.0020
−0.0025
−0.0030
−0.0035
−0.0040
$(E - E_0)/|E_0|$
0
200
400
600
800
1000
t/T_0
Leapfrog
RK4